

NTU Racing—Simple Electronic Tasks

I. Task 1

According to the MSP300 datasheet provided in the description, M3041-000005-05KPG identifies a gauge-pressure sensor with a measurement range of 0 to 5000 psi, a nominal analog output of 1 to 5 V, and a required supply voltage of 10 to 30 V. Since the MCU input must not exceed 3.3 V, we need a voltage divider. A suitable one can be constructed using a 10 k Ω resistor between the sensor output and the MCU input, and a 15 k Ω resistor between the ADC input and ground, as shown in the diagram, giving a division ratio of 0.6.

The divider converts the sensor's 1~5 V output to 0.6~3.0 V, leaving a safe margin below the 3.3 V limit. It draws only 40~200 μ A and presents a total load resistance of 25 k Ω , satisfying the sensor's load requirement. Both resistor values are widely available. If substitutes are required, the lower resistor should remain approximately 1.5 times the upper resistor. Excessively small values should be avoided because they increase current consumption, while excessively large values may affect ADC sampling accuracy. The sensor and MCU must share a common ground, and additional filtering may be added to reduce electrical noise and protect against voltage transients.

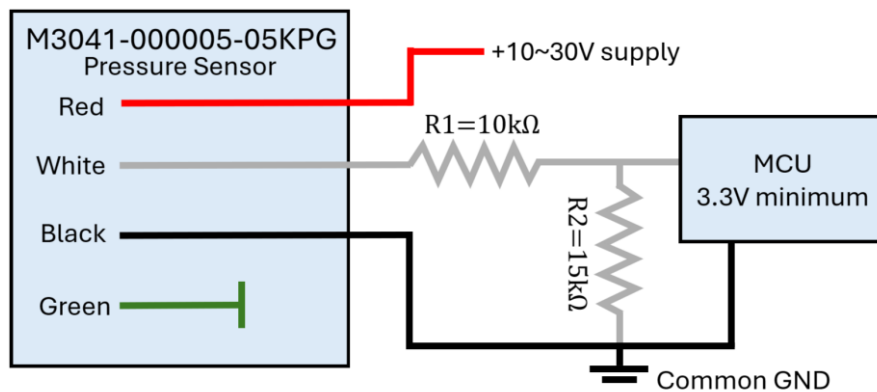


Figure 1. Voltage divider circuit

II . Task2

For the single QTLP690C red LED, use a 2N2222A as a low-side switch, connecting the 5 V supply through a 150 Ω , 0.25 W resistor and the LED to the collector, the emitter to ground, and the 3.3 V MCU output through a 1 k Ω resistor to the base. If we assume the QTLP690C-R has $V_F = 2.0$ V and $V_{CE} \approx 0.2$ V, then the required resistance must be at least

$$R_2 \geq \frac{5 - 2.0 - 0.2}{0.02} = 140\Omega$$

Hence, picking R_2 as 150 Ω is a reasonable choice with a more common value. Then, the current drawn to the LED becomes

$$I_{LED} = \frac{5 - 2.0 - 0.2}{150} \approx 18.7mA$$

And dissipating power around 0.052W. For the base resistor, use a forced current gain of approximately 10 and assuming $V_{BE} \approx 0.8$ V, we get R_1 should be smaller than 1.25k Ω . Hence, picking $R_1 = 1$ k Ω and a pull-down resistor $R_3 = 100$ k Ω is an available choice. For which the pull-down resistor only takes an 8 μ A current and the actual base current is approximately 2.492mA. The transistor dissipates approximately 5.7 mW and has an estimated temperature rise of about 1.1 $^{\circ}$ C, so the design safely satisfies the MCU current, transistor current, voltage, power and temperature requirements. Figure 2 demonstrates the circuit satisfying the requirements I found during the analysis.

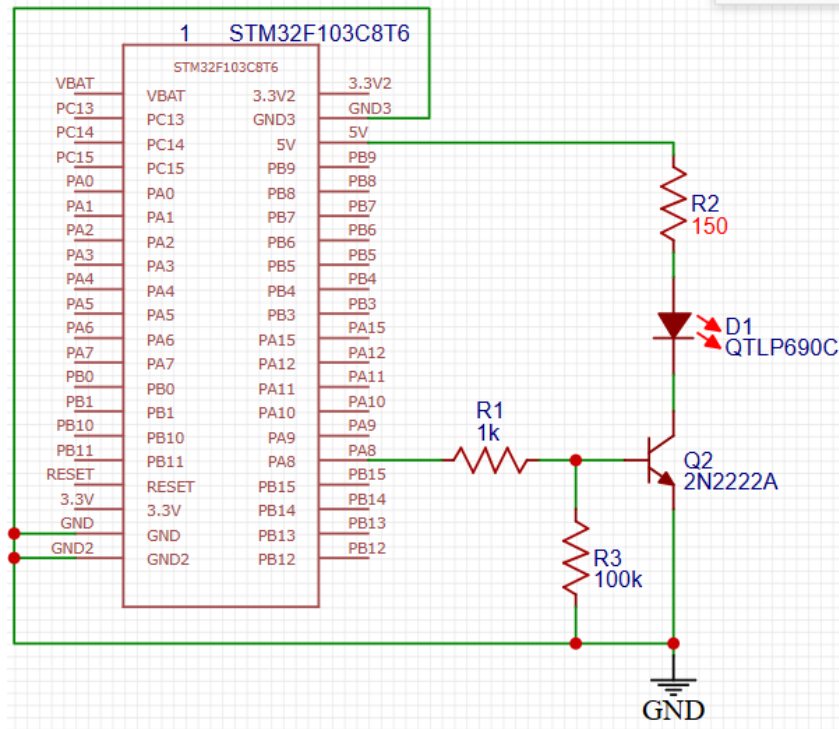


Figure 2. Circuit for a single LED

For the 20-LED array, arrange the LEDs as 20 parallel branches, each with its own $150\ \Omega$, $0.25\ \text{W}$ resistor, giving a total load current of approximately $400\ \text{mA}$, as calculated below

$$I_{LED} = \frac{5 - V_F}{150 + 20R_S} = \frac{5 - 2}{150 + 20 \times 0.03} \approx 19.9\text{mA}$$

$$I_{total} = 20I_{LED} \approx 398\text{mA}$$

I use an IRLR2908 N-channel MOSFET as the low-side switch, with its source connected to ground, its drain connected to the common LED bus, its gate driven through a $220\ \Omega$ resistor, and a $100\ \text{k}\Omega$ gate-to-source pull-down resistor.

The $220\ \Omega$ resistor limits the initial gate-charging current to at most $\frac{3.3}{220} = 15\ \text{mA}$, while the MOSFET draws almost no steady-state gate current. The IRLR2908 is the best of the four choices because it is a logic-level MOSFET with very low on-resistance and therefore has negligible voltage drop, power loss, and temperature rise at $400\ \text{mA}$. Each LED must have an individual resistor to prevent uneven current sharing.

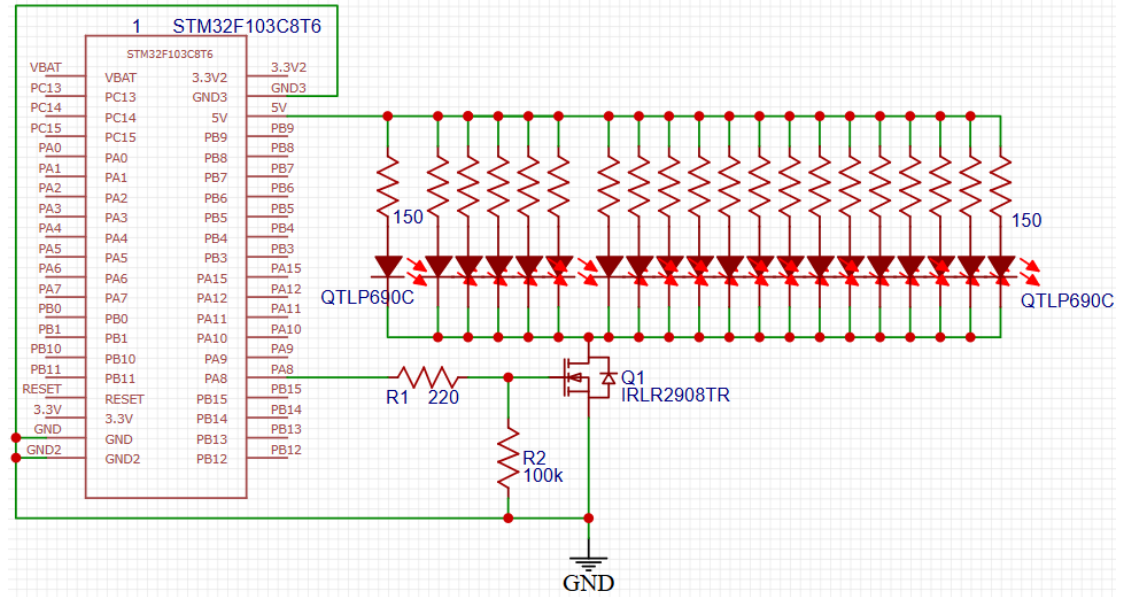


Figure 3. Circuit for 20-LED array